F FWallSim · Firewall Course

CLASS 17

PC2 has no network

Troubleshooting: DHCP

Advanced | ~10 minutes | FWallSim v1.0.0 (FortiOS 7.4 style)

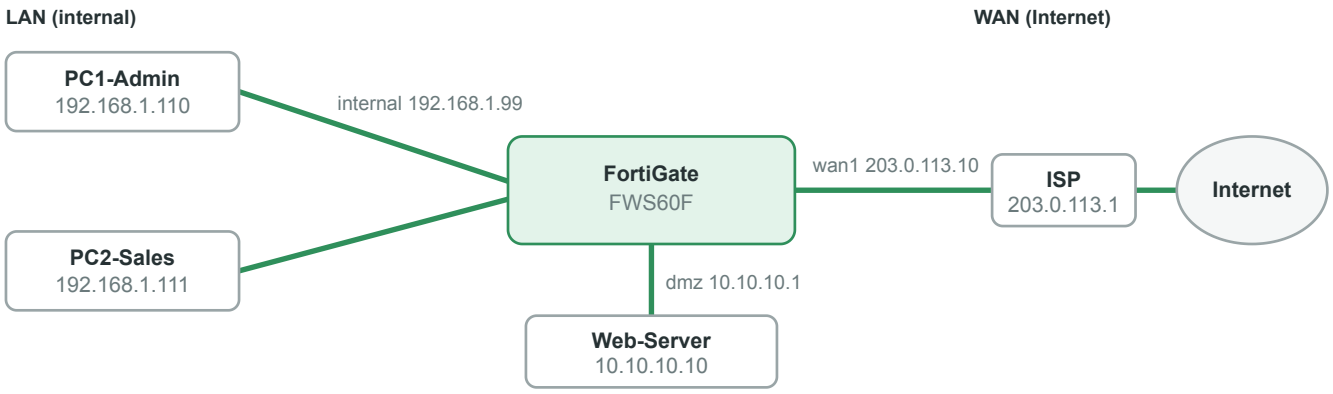
YOU WILL LEARN

- Troubleshooting: DHCP

BEFORE YOU START

Finish Class 16 first.

The lab network



The goal

THE IDEA

PC1 works, but PC2 has no internet at all after a reboot.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **PC2 has no network** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1 **PC2 has an address and can browse**

Check PC2 in the Traffic Generator (its address is shown under Source).

Hints

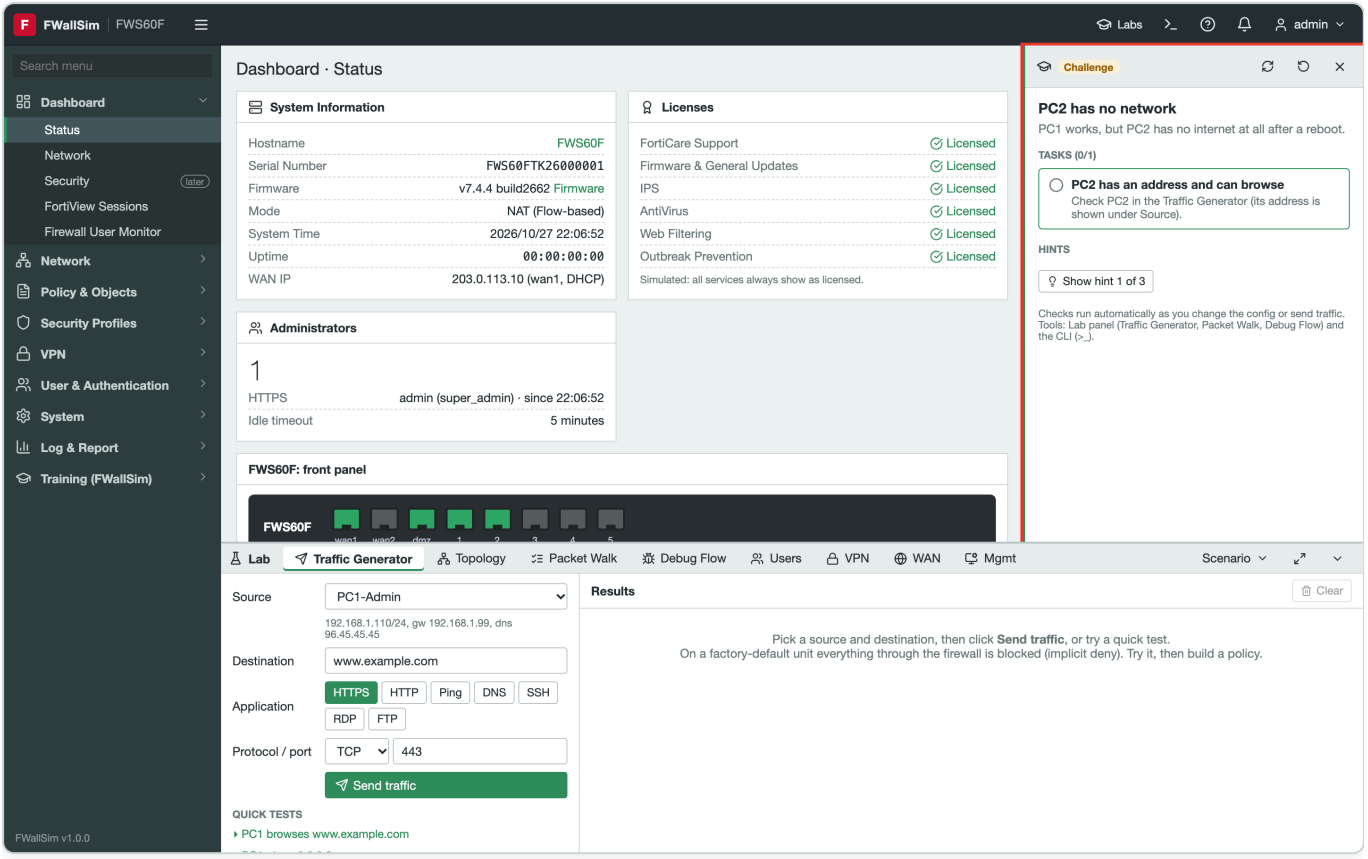
1. What IP does PC2 have? 169.254.x.x means DHCP failed.
2. Dashboard > Network > DHCP shows the leases.
3. How many addresses does the DHCP range contain?

Solution

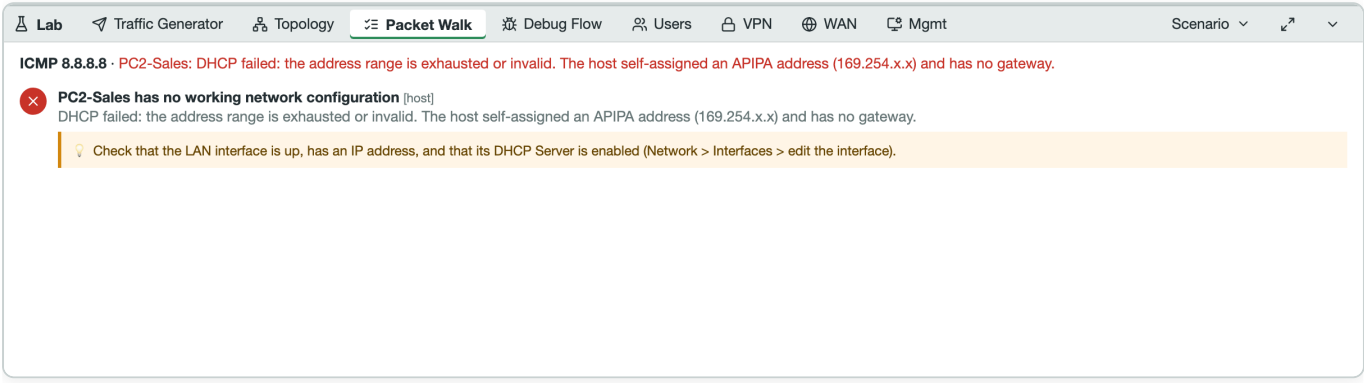
1. Network > Interfaces > edit internal.
2. DHCP Server address range: 192.168.1.110 - 192.168.1.210.
3. OK, then test again.

Screenshots

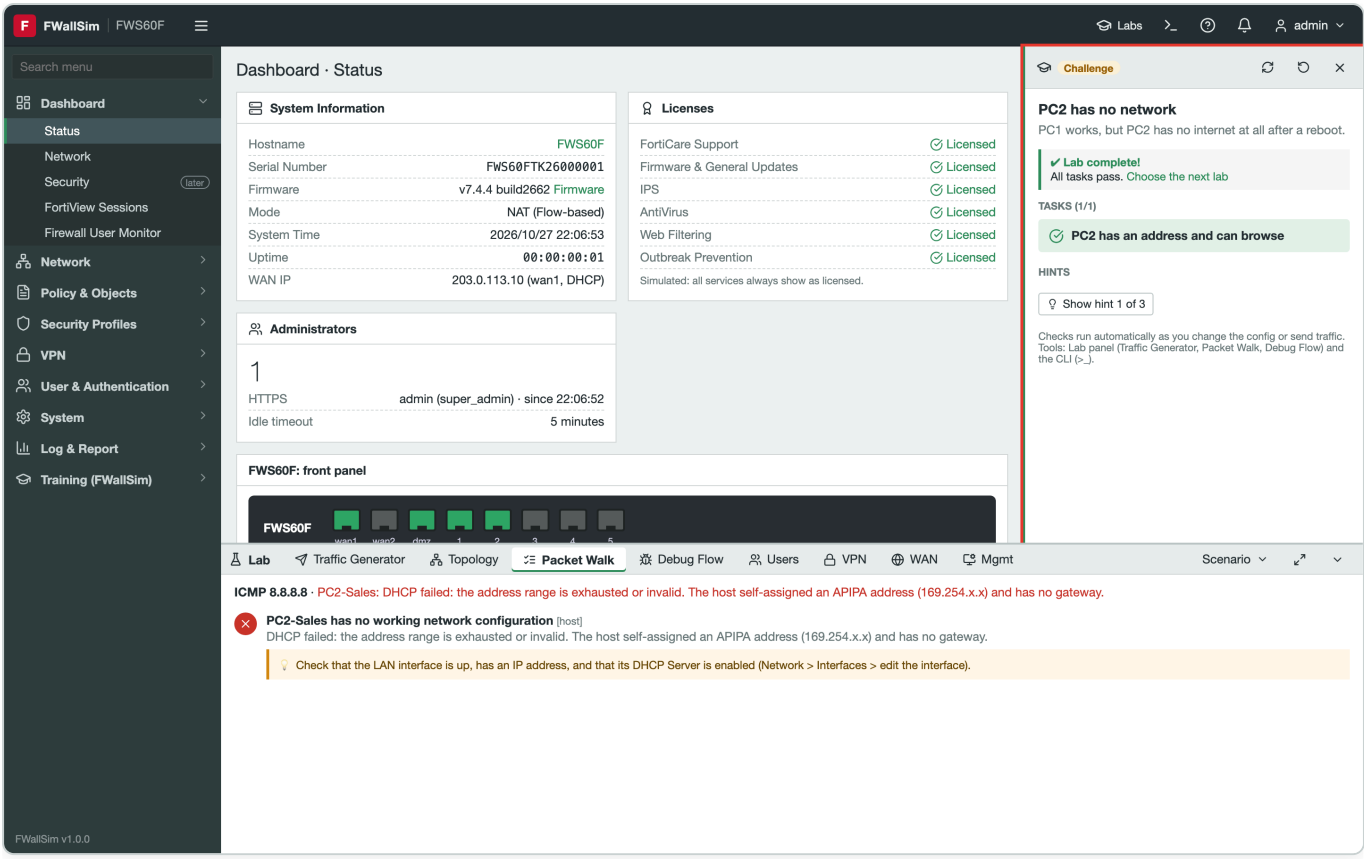
Numbered red markers show where to click or what to look at.



Lab Started



Packet Walk Before



Lab Solved

Interfaces

+ Create New | Edit | Delete | Search

Name	Type	Members	IP/Netmask	Role	Administrative Access	DHCP Clients
Physical Interface (3)						
dmz	Physical Interface		10.10.10.1 255.255.255.0	DMZ	PING, HTTPS, Security Fabric Connection	0
wan1	Physical Interface		203.0.113.10 255.255.255.0 (DHCP)	WAN	PING, FMG-Access	0
wan2	Physical Interface		0.0.0.0 (DHCP: no lease)	WAN	PING, FMG-Access	0
Hardware Switch (1)						
internal	Hardware Switch	internal1 internal2 internal3 internal4 internal5	192.168.1.99 255.255.255.0	LAN	PING, HTTPS, SSH, Security Fabric Connection	2
Tunnel Interface (1)						
ssl.root (SSL VPN interface)	Tunnel Interface		0.0.0.0 0.0.0.0	Undefined		0

Tip: double-click an interface to edit it. Interfaces used by policies, routes or DHCP servers show a Ref. count and cannot be deleted until those references are removed.

Interfaces

Dashboard · Network

Static & Dynamic Routing (equivalent of: get router info routing-table all / database)

Network	Gateway IP	Interfaces	Distance	Priority	Type	Status
10.10.10.0/24	0.0.0.0	dmz	0		C Connected	Active
192.168.1.0/24	0.0.0.0	internal	0		C Connected	Active
203.0.113.0/24	0.0.0.0	wan1	0		C Connected	Active
0.0.0.0/0	203.0.113.1	wan1	5	1	S* Static (DHCP)	Active

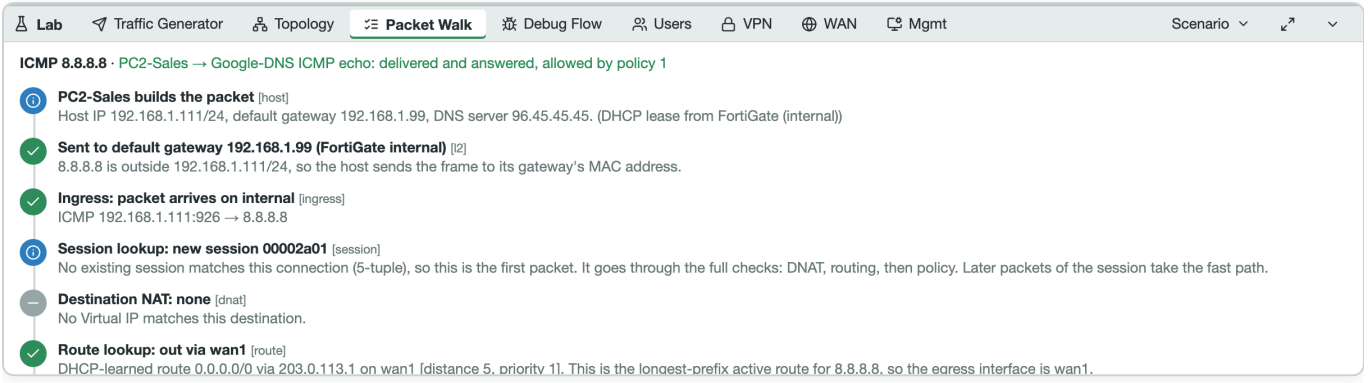
DHCP

Device	IP Address	Interface	Gateway / DNS handed out	Status
PC1-Admin	192.168.1.110	internal	192.168.1.99 / 96.45.45.45	Leased
PC2-Sales	192.168.1.111	internal	192.168.1.99 / 96.45.45.45	Leased

Interface traffic (since boot)

Interface	Link	IP	RX bytes	TX bytes	RX packets	TX packets
dmz	up	10.10.10.1/24	0 B	0 B	0	0
wan1	up	203.0.113.10/24	0 B	0 B	0	0
wan2	no link	–	0 B	0 B	0	0
internal	up	192.168.1.99/24	0 B	0 B	0	0
ssl.root	up	–	0 B	0 B	0	0

Dashboard Network



Packet Walk After

Next: Class 18: The branch office is cut off